CBOR can be parsed and processed without a schema. On top of CBOR, the CBOR Object Signing and Encryption (COSE, https://tools.ietf.org/html/rfc8152) standards provide similar cryptographic capabilities as JOSE does for JSON.

> **DEFINITION *CBOR*** (Concise Binary Object Representation) is a binary alternative to JSON. ***COSE*** (CBOR Object Signing and Encryption) provides encryption and digital signature capabilities for CBOR and is loosely based on JOSE.

Although COSE is loosely based on JOSE, it has diverged quite a lot both in the algorithms supported and in how messages are formatted. For example, in JOSE symmetric MAC algorithms like HMAC are part of JWS (JSON Web Signatures) and treated as equivalent to public key signature algorithms. In COSE, MACs are treated more like authenticated encryption algorithms, allowing the same key agreement and key wrapping algorithms to be used to transmit a per-message MAC key.

In terms of algorithms, COSE supports many of the same algorithms as JOSE, and adds additional algorithms that are more suited to constrained devices, such as AES-CCM and ChaCha20-Poly1305 for authenticated encryption, and truncated version of HMAC-SHA-256 that produces a smaller 64-bit authentication tag. It also removes some algorithms with perceived weaknesses, such as RSA with PKCS#1 v1.5 padding and AES in CBC mode with a separate HMAC tag. Unfortunately, dropping support for CBC mode means that all of the COSE authenticated encryption algorithms require nonces that are too small to generate randomly. This is a